

Draft main-text Results and Discussion

Predictability and genomic organisation of ancestry sorting

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Working draft excluding the epistasis screen and simulations

The first two sections are written as candidate manuscript text. The final section is an editorial figure and supplementary-material plan. Climate associations are interpreted as conditional and exploratory.

Results

Ancestry sorting is predominantly directional across hybrid populations

To determine whether hybrid genome evolution repeatedly favoured the same parental ancestry, we quantified near-fixation across twenty geographically separated *F. aquilonia* $\times$ *F. polyclena* populations. The analysis extends the three-population comparison of Nouhaud et al. (2022) in two respects: it separates repeated fixation in the same parental direction from fixation in opposing directions, and it accounts for local linkage rather than assigning equal weight to fixed physical windows.

Among SNPs that remained polymorphic in the pooled parental samples, 67.96% were not near-fixed in at least half of the hybrid populations. Of those that did show repeated sorting, 17.50% sorted toward *F. polyclena* and 14.40% toward *F. aquilonia*, whereas only 0.14% showed strong sorting in opposing directions among populations (Fig. 1A). Thus, the main feature of the data was not near-fixation itself, but the consistency of its parental direction across populations. Because the populations may share aspects of their demographic history, these proportions describe repeatability across the sampled panel rather than proving independent evolutionary replication.

Marker-level summaries can overstate the genomic extent of repeated sorting when many correlated markers report the same event. We therefore reduced 1,114,340 markers to 474,014 LD clusters and represented each cluster once. Among clusters selected because they contained at least one SNP-level sorting signal, aggregation removed the sorting classification increasingly often as cluster size increased: 74.7% of clusters containing at least five markers were unsorted after aggregation, compared with 5.0% of the smallest units. This comparison is conditional on the presence of a marker-level signal and does not estimate the genome-wide fraction of unsorted clusters. The loss of the aggregate call was unlikely to result primarily from poor consensus construction, because consensus fidelity remained high (median 0.856) and only 7 of 16,597 multi-marker unsorted clusters fell below the predefined fidelity threshold. Repeated sorting was therefore distributed mainly across numerous small genomic units rather than a limited set of large LD regions.

The parental direction of sorting changes along the diagnostic-index axis

The modest genome-wide excess of sorting toward *F. polyclena* concealed a strong reversal in direction along the diagnostic-index axis. Among unidirectionally sorted loci, the fraction sorting

toward *F. aquilonia* increased monotonically from approximately 0.15 in the least diagnostic decile to 0.74 in the most diagnostic decile, crossing parity near the centre of the DI distribution (Spearman $\rho = 0.31$; Fig. 1B). In a logistic model including pooled parental MAF, DI strongly increased the probability of *F. aquilonia*-directed sorting (DI coefficient = 0.082, $z = 181$), whereas higher parental MAF weakly shifted sorting toward *F. polyclena*.

Larger LD clusters were both more diagnostic and more likely to sort toward *F. aquilonia* in unadjusted summaries. This relationship did not reflect an independent effect of cluster size. After DI and parental MAF were included, the size coefficient reversed direction ($\log_2(\text{size}) = -0.20$, $z = -22$), whereas the DI coefficient remained nearly unchanged (0.083, $z = 120$). Parental diagnosticity, rather than the physical extent of a correlated region, was therefore the principal axis organising the direction of repeated sorting.

Linkage changes the apparent abundance, but not the main direction, of sorting

DI was nearly orthogonal to recombination rate ($\rho = -0.03$) and pooled parental MAF ($\rho = -0.02$). At the analysed SNPs, high DI was associated most strongly with lower parental expected heterozygosity ($\rho = -0.46$) and only weakly with the between-parent allelic-difference summary ($\rho = 0.13$). The diagnostic-index gradient therefore cannot be reduced to a low-recombination gradient or to elevated between-parent difference alone.

The importance of representing LD-correlated markers once was clearest in regions of lowest recombination. In the lowest recombination decile, 17% of SNPs were classified as sorted, but only 6% of LD-reduced units were sorted, compared with approximately 42% of units across the remaining deciles (Fig. 1C). The excess marker-level signal in low recombination therefore largely reflected spatial pseudoreplication. Consistent with this interpretation, aggregate sorting increased rather than decreased with recombination rate.

Among unidirectionally sorted units, DI remained by far the strongest predictor of parental direction in a model including recombination, parental MAF, and cluster size (standardised DI coefficient = 1.46, $z = 125$). Recombination made a much smaller contribution (-0.09 , $z = -7.6$), corresponding to a weak tendency toward *F. aquilonia* ancestry at lower recombination rates. This residual association is descriptive and is not sufficient on its own to identify its evolutionary cause. Nevertheless, both relationships were robust to residual genomic dependence. A chromosome block bootstrap gave 95% intervals of 1.41–1.52 for the DI coefficient and -0.145 to -0.037 for the recombination coefficient; intervals from 10-cM genetic-map blocks were similar. The positive relationship between recombination and sorting magnitude was likewise retained (coefficient = 0.052, chromosome-block 95% interval 0.046–0.057).

Climate association does not robustly explain diagnosticity or repeated directional sorting

The hybrid populations span marked climatic variation. The first two principal components of nineteen bioclimatic variables jointly explained 86% of this variation. PC1 primarily captured a temperature and seasonality gradient, with the high-altitude Swiss population resembling colder northern populations, whereas PC2 primarily separated populations by precipitation. Genotype–environment associations were evaluated while accounting for among-population covariance; because PC2 covaried with genome-wide ancestry and this relationship was strongly influenced by Åland, the Åland-excluded analysis with a fixed covariance matrix was treated as primary.

Under model-based uncertainty, climate-associated LD clusters appeared to contain a higher proportion of ancestry-diagnostic markers. Among clusters with at least fifty markers, a 10-

percentage-point increase in the proportion of climate-associated members corresponded to increases of 4.5 percentage points for PC1 ($P = 8 \times 10^{-4}$) and 3.7 percentage points for PC2 ($P = 0.013$) in the proportion of diagnostic markers. This nominal enrichment was not robust to genomic dependence: chromosome block-bootstrap 95% intervals were -4.0 to 13.1 for PC1 and -0.7 to 17.8 for PC2, with similar conclusions from 10-cM blocks (Fig. 1D).

The estimates were also attenuated after accounting for variables related to allele-frequency variation and power. Adding mean hybrid MAF reduced the PC1 and PC2 coefficients to 3.5 and 2.5 percentage points, respectively. After further adjustment for mean recombination rate and BayPass XtX, a measure of among-population differentiation, the corresponding coefficients were 1.8 and 1.7 percentage points ($P = 0.16$ and 0.23); chromosome block-bootstrap intervals again included zero. Because XtX and the climate Bayes factor are derived from the same allele-frequency data, this is a conservative sensitivity analysis rather than an estimate of an independently adjusted climate effect. Together, the analyses show that the nominal enrichment is compatible with shared dependence on allele-frequency differentiation and cannot be identified as a climate-specific association.

The same climate-association rate did not predict repeated directional sorting. In clusters with at least fifty markers, the odds ratios were 0.99 for PC1 and 1.03 for PC2, with $P > 0.95$ for both. This absence of enrichment should be interpreted cautiously: genotype–environment association requires allele-frequency variation among populations, whereas repeated sorting toward the same parent removes that variation. Nevertheless, the contrast is informative: the measured climatic gradients did not robustly explain either diagnostic content beyond general differentiation or the repeated direction of ancestry sorting.

Discussion

Across twenty hybrid populations, ancestry sorting was strikingly asymmetric: loci that repeatedly approached fixation almost always did so toward the same parent, while strong fixation in opposing directions was rare. This extends the repeatable sorting described across three populations by Nouhaud et al. (2022) to a much broader population panel. The additional populations also allow repeatability to be decomposed explicitly into consistent and opposing directions, although we remain cautious about treating all populations as fully independent evolutionary replicates.

The central result is not a uniform bias toward one parent. Instead, the parental direction of sorting reverses along the diagnostic-index axis: low-DI loci tend to sort toward *F. polychrysa*, whereas high-DI loci tend to sort toward *F. aquilonia*. This explains why analyses restricted to strongly diagnostic loci can yield an apparent *F. aquilonia* excess while the broader marker set shows a slight *F. polychrysa* excess. DI is not simply a proxy for recombination or cluster size. At the analysed SNPs it is associated mainly with reduced parental expected heterozygosity, suggesting that the direction of sorting is organised by the parental frequency landscape rather than by a single class of large, low-recombination regions.

Accounting for LD changes the biological picture. Marker-level analyses make sorting appear concentrated in low-recombination regions because many nearby markers repeat the same ancestry signal. Once each LD cluster contributes once, sorting is least frequent in the lowest-recombination decile and is carried primarily by numerous small units. The weak residual association between low recombination and *F. aquilonia*-directed sorting is therefore secondary to the much stronger diagnostic-index gradient. More generally, this comparison shows that the apparent genomic extent of repeatable evolution can depend strongly on whether markers or correlated genomic units are counted.

Climate association did not provide a robust explanation for repeated sorting. Directionally sorted units were not enriched in climate-associated regions, and the nominal enrichment of ancestry-diagnostic markers disappeared once residual genomic dependence was respected. Its attenuation after accounting for hybrid MAF and XtX further indicates that it could arise from the allele-frequency differentiation that contributes to both diagnosticity and BayPass evidence. XtX is a deliberately strong control and may remove biologically relevant as well as confounding variation, so this sensitivity analysis does not demonstrate that climate is irrelevant. Moreover, locally varying selection could drive populations in different directions, whereas repeated near-fixation reduces the among-population variation used to detect genotype–environment association. The present data nevertheless provide no robust evidence that the measured climatic gradients organise the repeated direction of ancestry sorting.

Together, the results identify diagnosticity as the main organising axis of early hybrid genome evolution in this system. Recombination determines how many markers repeat a signal and makes smaller but block-robust contributions to sorting magnitude and direction. In contrast, neither diagnostic content nor repeated directional near-fixation showed a robust climate-specific association. The emerging picture is therefore one of many small ancestry units sorting in a highly structured parental direction, rather than a few large genomic regions responding uniformly to linkage or the measured climatic gradients. These conclusions remain associative: shared population history and conditional testing limit causal interpretation, and individual candidate regions require independent biological scrutiny.

Editorial figure plan

Recommended main figure: one integrated ancestry-sorting figure

The main figure should carry the central narrative rather than the validation machinery. A four-panel figure is sufficient:

This layout is realised in Figure 1 (shown in the Results), which progresses from the phenomenon (A), to its principal predictor (B), to the effect of genomic linkage on its apparent abundance (C), and finally to its relationship with environmental association (D).

Panel A establishes the phenomenon. Panel B is the biological centre of the figure and should receive the most visual space. Panel C demonstrates why the LD reduction changes the interpretation. Panel D closes the extrinsic-climate branch without requiring separate Manhattan plots in the main text.

Material better placed in supplementary figures and tables

Supplementary item	Recommended content
LD decay diagnostics	Chromosome-specific decay fits, ROC evaluation of local LD support, and example genomic tracks.
Complexity-reduction validation	Stage 1 versus combined clustering in extended-LD regions, fidelity distributions, threshold sensitivity, residual inter-unit LD, and a concise parameter table.
Sorting robustness	Full class counts, conditional aggregation by cluster size, fidelity check for apparently diluted clusters, exact-test summaries, and sensitivity to sorting thresholds.
Genomic architecture	Full parental-summary and recombination decile plots, complete regression tables, and consensus-versus-representative validation.
Climate association	Ancestry–climate confounding check, nested cluster-size strata, the MAF/XtX sensitivity ladder, block-aware uncertainty, and full model tables.
Dependence-aware uncertainty	Chromosome- and 10-cM-block confidence intervals for the principal sorting, architecture, and climate coefficients, with the reconstruction checks and bootstrap diagnostics.

Suggested order in the full paper

The co-authors' sampling, ancestry, and climate overview should establish the system first. The text drafted here can then enter as one continuous section: (1) directional repeatability, (2) diagnostic-index polarity, (3) genomic architecture after LD reduction, and (4) overlap with climate association. This order lets the main figure function as a bridge from the descriptive population overview to the later mechanistic analyses without exposing the internal analysis-module structure.

Reference

Nouhaud P, Martin SH, Portinha B, Sousa VC, Kulmuni J. Rapid and predictable genome evolution across three hybrid ant populations. *PLoS Biology*. 2022;20:e3001914. <https://doi.org/10.1371/journal.pbio.3001914>.

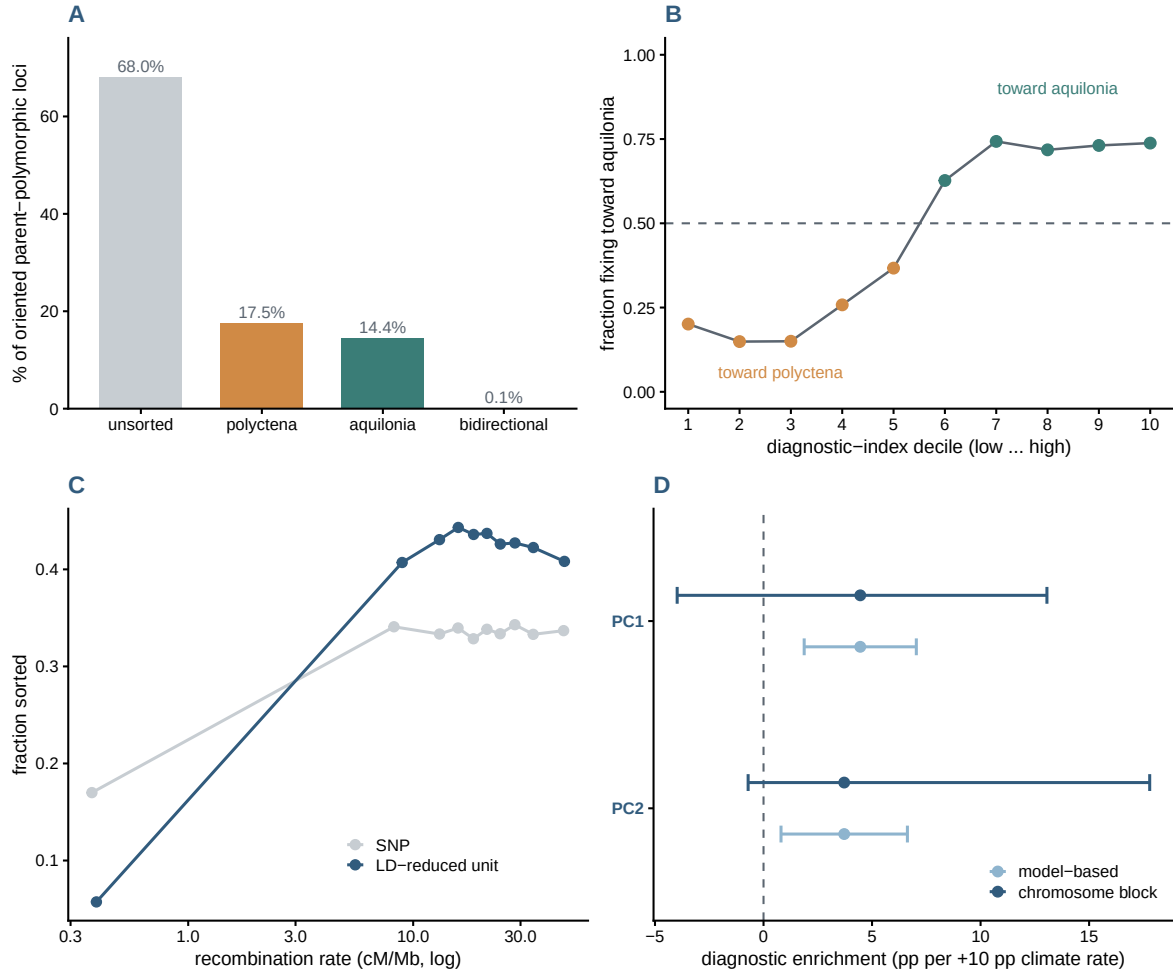


Figure 1: **Predictability and genomic organisation of ancestry sorting.** (A) Sort-class proportions among oriented parent-polymorphic loci: near-fixation is predominantly unidirectional, with bidirectional fixation almost absent. (B) Among unidirectionally sorted loci, the fraction fixing toward *F. aquilonia* rises across diagnostic-index deciles, crossing parity near the centre. (C) Fraction sorted versus recombination at the SNP and LD-reduced-unit levels; the unit-level collapse in low recombination reflects removal of spatial pseudoreplication. (D) Diagnostic-index enrichment of climate-associated clusters (percentage points per +10 percentage-point climate rate) for PC1 and PC2, with model-based and chromosome block-bootstrap 95% intervals; the nominal enrichment is not robust to genomic dependence.